

... and pigs will fly

IRON MAN IS AT IT AGAIN. Elon Musk held a press event for Neuralink, the startup that he founded to connect brains with computers in a symbiotic relationship. Neuralink even had a live demonstration with three pigs of which one had a Neuralink inserted, one had it removed and was supposedly healthy and well, and the last one was just a control which had not been given a Neuralink. The device is extremely small and sits flush with the skull once it has been installed in place. There are multiple links emanating from the device with 1024 channels per link. This is a magnitude higher than what's currently available in the market. Musk describes Neuralink as a Fitbit in your skull. It's quite amazing how Elon Musk jumps from one industry to another and manages to completely accelerate the technological progress in a fraction of the time compared to the industry incumbents.

The potential applications for Neuralink are enormous. Anything that can be helped or solved by stimulating sections of the brain is a potential application for Neuralink. This includes the numerous mental health issues that plague millions of human beings. Musk's presentation highlighted several neurological problems such as memory loss, addiction, anxiety, strokes, paralysis, blindness, depression, insomnia and many other problems for which Neuralink could be applicable. Now, it should be made clear that there are numerous ways in which any of the neurological problems listed above can manifest in a person and while Neuralink can potentially address some of these, it shouldn't be considered a panacea. One can certainly hope that it does surpass all expectations but let's be grounded till we see things actually happen.

Most of the problems listed above can be addressed by stimulating sections of the brain and reading the response to those stimuli. If you extend this read/write methodology, one might be able to record and view memories. It's a really long shot but once you map the firing patterns of neurons to real world activities, you can potentially link up with a computer and store or retrieve memories. Take this concept one step further and you can have Cortical Stacks like they do

in Altered Carbon. We would be able to move from one body to another. We're already growing individual body parts in the lab as a means of natural prosthetics, put the two together and we'll have cloning stations that can create replicates without the long wait of actually growing clones. As cool as it sounds, the concerns around these grow just as quickly. One need not look far since the very same sci-fi shows do an excellent job of showcasing the downsides as well.

Neuralink's long-term goal is to achieve symbiosis with artificial intelligence. If the simple goal of reading and storing memories for later has you a little spooked, then AI symbiosis should certainly send you over the edge. Skynet no longer needs to fight humans if it can simply tack little chips onto their heads and make them a subservient army of robots. We're ages away from even building an Artificial General Intelligence (AGI) but the thought of AI meddling with your brains does make a cause for concern. Implementing AI in any domain rapidly accelerates the pace of development and AI can do the same with Neuralink as well. The data read from your brain has to be interpreted properly and an AI trained to do that will lead to quicker results, so it's inevitable that an AI will be used sooner than later with Neuralink. It just won't be Skynet or anything even close to being as smart.

Like all companies, Neuralink is also a business. And that too in the medical space. Musk has an uphill task of breaking into the space since not everyone accepts his eccentricity. Some researchers find it a mockery because Neuralink is doing rudimentary work that has been done thousands of times in the past and getting all the limelight. Whereas cutting edge research is getting defunded on a daily basis. Then again, Musk is taking existing technology and polishing and making it better far quicker than these researchers can. The way Musk has pulled off miracles with SpaceX and Tesla does tell us that Neuralink has immense potential and it will step on the toes of some of the biggest players in the medical space but I feel this is necessary since the problems that Neuralink can alleviate will improve millions of lives and save just as many. **d**



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